

Objectives

To implement a distributed dropout risk prediction framework using Apache Spark by applying a rule-based heuristic scoring model and generating performance metrics based on student risk classifications. This phase also aims to develop comprehensive data visualizations that transform large-scale analytical results and machine learning outputs into clear, interpretable, and decision-oriented insights. Through the integration of performance analysis and visual reporting, the activity seeks to highlight the relationships between academic and psychological indicators and student dropout risk, facilitating effective communication of findings and supporting data-driven decision-making.

Tasks Completed

- Cleaned, preprocessed, and normalized 1,000,000 student records using distributed Spark RDD operations to ensure consistency and readiness for large-scale analysis.
- Applied a rule-based heuristic scoring algorithm by assigning weighted contributions to risk and protective factors, then segmented computed scores into Low, Medium, and High Risk tiers using percentile thresholds (33rd and 66th).
- Cross-validated predicted risk labels against ground truth data to generate baseline accuracy metrics and construct a multiclass confusion matrix for classification performance assessment.
- Developed key data visualizations, including correlation heatmaps, risk distribution charts, feature importance analysis, and confusion matrix heatmaps to clearly communicate relationships between psychological factors and dropout risk.

Tasks to be Completed

N/A

Results and Analysis

- The model produced an imbalanced classification distribution of 76.7% Low Risk, 21.8% Medium Risk, and 1.5% High Risk, where a naïve baseline of predicting all students as Low Risk would already yield 76.7% accuracy; however, the multiclass confusion matrix provided a more meaningful assessment by capturing true positives within the minority High Risk group.
- Feature analysis identified burnout (+1.5) and stress (+1.4) as the strongest drivers of high-risk classification, indicating that psychological exhaustion has a greater impact on dropout risk than standard academic pressure.
- Visualization results effectively translated model outputs into interpretable insights, showing that severe psychological strain increases dropout risk, while strong mental health and social support serve as key protective factors.

Challenges and Solutions

A key challenge encountered was that overall accuracy metrics were misleading due to the highly imbalanced dataset, where 76.7% of students were classified as Low Risk, masking the model's effectiveness in identifying the 1.5% High Risk group. This was addressed by focusing on true positive density within the minority class using a multiclass confusion matrix to better reflect actual detection performance. Another challenge involved visualizing a large-scale dataset of 1,000,000 records, where direct plotting could lead to memory issues and cluttered outputs. This was resolved by leveraging Apache Spark's distributed aggregation functions, such as `reduceByKey` and `collectAsMap`, to compute summarized statistics at the cluster level. Only aggregated results, including distribution counts and matrix summaries, were transferred to the local environment for visualization, ensuring efficient, stable, and clear rendering using Matplotlib and Seaborn.

Conclusion

The study showed that big data evaluation is more effective using multiclass confusion matrices rather than global accuracy, especially in imbalanced datasets. The pipeline successfully identified the 1.5% high-risk group, enabling more precise performance assessment and supporting targeted intervention strategies. The objective was also achieved through the creation of clear statistical visualizations that simplified complex patterns in the student mental health and burnout dataset. These visual outputs improved interpretability, connected data processing with decision-making, and prepared the project for final presentation and deployment.